Prototype Roadmap — Commission Research Intelligence Dashboard

Living document. Updated as features are built, tested, or planned.

Feature Inventory

Feature	Status	Notes
Data Pipeline		
Transcript download (Supabase API)	✅ Complete	`madlanga_downloader.py` — fetches all hearings, media, documents
Transcript parsing (.txt → segments)	✅ Complete	`parse_transcripts.py` — speaker-tagged segments
Statement parsing (PDFs → text)	✅ Complete	`parse_statements.py`
Speaker normalization	✅ Complete	`normalize_speakers.py` — raw labels → canonical names + roles
Entity extraction (regex + gazetteers)	✅ Complete	`extract_entities.py` — 100k+ entity mentions, 16k+ unique
FTS5 search index	✅ Complete	`index_search.py` — SQLite full-text search
Network computation (bipartite graph)	✅ Complete	`compute_network.py` — person→entity edges with accusation scoring
Story map construction	✅ Complete	`build_story_map.py` — dual timeline (documents + testimony)
Dashboard		
Overview — stat cards + mini network	✅ Complete	Top 40 persons + 40 entities, concentric layout (static)
Connections — full network graph	✅ Complete	Filterable by connection type, concentric layout (static)
Witnesses & Entities — role-based cards	✅ Complete	Speaker profiles with entity mentions
Attention Map — questioning heatmap	✅ Complete	Who questioned whom, intensity matrix
Timeline — hearing chronology	✅ Complete	Interactive plot + table of all hearing days

Feature	Status	Notes
Search — FTS5 full-text	✅ Complete	Keyword/phrase/wildcard/boolean, speaker + date filters
Entities — entity browser	✅ Complete	16k+ entities browsable by type, context snippets
Story Map — parallel timelines	✅ Complete	Docs vs testimony, entity + witness filters
Evidence Compare — side-by-side	✅ Complete	What documents say vs what witnesses say about any entity
Contradictions — flagged conflicts	✅ Complete	Entities mentioned by multiple witnesses
Dashboard — stats page	✅ Complete	Summary cards, top accused, evidence corroboration, validity breakdown
Concentric Network Layout	✅ Complete	Witnesses on inner circle, entities on outer circle, barycenter ordering. TODO: re-enable physics for drag interaction.
Planned / In Progress		
Evidence Bundle Export	⚠️ Partial	All data exists — needs UI to format as printable HTML/PDF
Witness Preparation Pack	🔲 Planned	One-click witness summary card
Contradiction Engine (dedicated view)	🔲 Planned	Auto-flag conflicting testimony across witnesses
Multi-user annotations	🔲 Planned	Shared notes on excerpts
Citation export with line references	🔲 Planned	“Day 47, p.23, lines 10-15” format
Student research feedback integration	🔲 Planned	Incorporate findings from law student research brief

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Status Legend

Icon	Meaning
✅ Complete	Built, loaded with real data, tested, deployable
⚠️ Partial	Functional but needs polish before client demo
🔲 Planned	Designed but not yet implemented

Recent Changes

Date	Change
2026-07-25	Concentric layout applied to Overview and Connections networks
2026-07-25	Network graphs set to static mode. TODO in code to re-enable physics.
2026-07-25	Value Analysis rewritten — new time-recovery framing + delay cost data
2026-07-25	Student Research Brief created for law student workflow research
2026-07-25	Investigative Story Package — 3 worked examples for journalists
2026-07-25	Auto-PDF watcher (`tools/md_to_pdf_watcher.py`) — professional styling via pandoc + wkhtmltopdf

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Notes

- The tool is **purpose-built for the Madlanga Commission** — not a generic platform
- All pipeline scripts are in `commission_research/pipeline/`
- All dashboard code is in `commission_research/app/`
- Features listed as “planned” should be informed by the Student Research Brief findings
- The student research brief at `research/Student Research Brief - Commission Workflow.md` outlines what needs investigation